

Adaptive Interview Questioning Rules (Literature-Informed)

- **Adaptive Difficulty:** If a candidate answers correctly and confidently, the system *escalates* difficulty; if they struggle, the system *de-escalates*. This follows standard adaptive-testing practice – as Cambridge explains, “if the student is doing well and answering questions correctly, the difficulty level of the questions increases. If or when [they] start struggling, the algorithm... gives them easier questions” [11†L158-L162] . (For moderate performance, difficulty can be held steady until a clear trend emerges.)
- **Probe Missing Concepts (Scaffolding):** If an answer omits key concepts, the next question should target those gaps. Educational tutoring theory insists that when prerequisites are not mastered, the tutor provides *guided, concept-targeted prompts* [23†L109-L113] . In practice, the interviewer might explicitly ask about a missing topic (e.g. “You mentioned X but not Y – can you explain Y?”). This mirrors a scaffolding strategy: first ask a question to *test understanding*, then offer support to fill gaps [36†L1518-L1521] . As one educational example puts it, “ask the child to [do a task] to discover...difficulties, then render support” [36†L1518-L1521] . In interviews, that means probing incomplete answers to reveal and address weak concepts.
- **Follow-Up Probing:** Even when an answer is partially correct, asking a related follow-up can deepen assessment. This is another form of scaffolding: one question leads to another to uncover subtler errors or uncertainties [36†L1518-L1521] . For instance, if a candidate’s reasoning seems good but they leave out details, the system can immediately follow up (e.g. “Could you elaborate on how X applies here?”). Such follow-ups help the interviewer identify misunderstandings in real-time, analogous to tutoring dialogues that incrementally test a learner’s knowledge [36†L1518-L1521] .
- **Application-Level Questions:** Once basic understanding is confirmed, test the candidate with an applied problem. In Bloom’s taxonomy, higher-level questions (analysis, application) require more domain knowledge than recall tasks [38†L132-L139] . In practice, that means after an explanatory answer, the interviewer asks an “apply this concept” question (for example, a real-world scenario or coding problem) to see if the candidate can transfer their knowledge. As one guide notes, “the higher the question in the hierarchy, the more domain knowledge the candidate needs to demonstrate” [38†L132-L139] , so matching question type to skill level is crucial.
- **Switch Topic after Mastery:** If a candidate repeatedly answers difficult questions in one topic correctly, it signals mastery and it’s time to move on. Mastery-learning theory holds that a learner should master key concepts *before* advancing [34†L84-L89] . Thus, if the candidate handles several hard OOP questions in a row, the next question should cover a new area (e.g. databases). This ensures broad coverage and prevents redundant questioning on already-mastered material [34†L84-L89] .

Each rule above is grounded in education and adaptive-testing literature, ensuring our synthetic interview dataset follows logical, evidence-based patterns. For example, Cambridge’s adaptive

assessment explicitly enforces the “harder-when-right, easier-when-wrong” logic 【11†L158-L162】 , while pedagogical research emphasizes scaffolding and focusing on unmastered concepts 【23†L109-L113】 【36†L1518-L1521】 . Together, these form a coherent rule set for generating realistic interview question sequences.

Sources: Principles of adaptive testing and tutoring (Cambridge Insight, 2023 【11†L158-L162】 ; Zhao *et al.*, *SHAPE 2026* 【23†L109-L113】) and educational theory on scaffolding and mastery learning 【36†L1518-L1521】 【34†L84-L89】 【38†L132-L139】 .
